

A Counterexample to Conjecture 00000007611

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Claim and interpretation

We use the standard Euclidean Clifford interpretation: a gamma system in dimension n is a family of n real matrices satisfying $\gamma_i^2 = I$ and $\gamma_i\gamma_j + \gamma_j\gamma_i = 0$ for $i \neq j$. The conjecture does not specify a coefficient field or the meaning of “dimension”; here the proposed lower bound is read as a lower bound on matrix size.

Counterexample

Take $n = 2$ and

$$\gamma_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \gamma_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Direct multiplication gives $\gamma_1^2 = \gamma_2^2 = I_2$ and $\gamma_1\gamma_2 + \gamma_2\gamma_1 = 0$. Hence these are a two-generator gamma system on a 2-dimensional vector space. The claimed lower bound would require matrix size at least $2n = 4$, contradicted by this example. The source also mentions tightness, a trace identity, and dimension counting without defining them; this counterexample addresses the explicit lower-bound clause only.

Formal check

The accompanying Lean 4 project verifies the displayed integer-entry matrix products and the numerical inequality $2 < 4$ without extra axioms. The integer entries are viewed as real entries in the mathematical example.